

VEDANT PATIL

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EDUCATION

K.J.Somaiya Institute of Technology, Mumbai

B.Tech in Electronics and Telecommunication and Honours in Data Science

CGPA: 9.47/10

Nov 2022 – May 2025

Trimurti Polytechnic

Diploma in Computer Engineering

Percentage: 80%

Dec 2020 – May 2022

CAREER OBJECTIVE

- Passionate software developer with expertise in Python, SQL, AI/ML and Full-Stack development, eager to apply technical skills in Data Science and software engineering to create innovative solutions.
- Experienced in building scalable applications and passionate about applying machine learning to solve real-world problems in collaborative, innovative environments.

WORK EXPERIENCE

Machine Learning Intern

Nov 2023 – Jan 2024

Kids Galaxy, Mumbai

- Developed an AI based chatbot to assist and guide students and help them to solve queries using chat services.

Web Development Intern

Jan 2024 – April 2024

Tantrasoft Solutions, Mumbai

- Assisted in developing and customizing client web applications using front-end technologies, including fixing bugs and implementing new UI features. Optimized database queries and improved front-end performance, reducing load times by 25%.

Full Stack Developer (Freelancer)

Jun 2024 – Sep 2024

IoT Cell, KJSIT, Mumbai

- Built an event management platform with online registration & Razorpay payment integration, digitizing 100% of fest entries and reducing manual work.

Data Science Intern

Nov 2024 – Feb 2025

Spinnaker Analytics, Mumbai

- Collaborated with the data team to clean, preprocess, and analyze large datasets, building ETL pipelines and creating visualizations in Matplotlib that improved business reporting efficiency by 25%. Assisted in developing and validating predictive models with scikit-learn, including feature engineering and tuning.

TECHNICAL SKILLS

Languages – Python, SQL, C++

Analytical – Machine Learning, Deep Learning (Neural Networks, CNN, LSTM), NLP, Statistical Modelling

Frameworks & Libraries – NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn, Django, Flask

Web Development – HTML5, CSS3, JavaScript, ReactJS, PHP, WordPress

Databases – MongoDB, MySQL

Tools – Git, Linux

PROJECTS

Early Breast Cancer Detection using Machine Learning and Deep Learning (Python)

- Built an interactive system to classify mammogram images (*dataset: 9,686 images*) as *Benign* or *Malignant* using 8 ML/DL models (SVM, Random Forest, XGBoost, CNN, etc.).
- Achieved 90%+ testing accuracy across models, with XGBoost (93.4%), Random Forest (93.1%) and CNN (91.9%) performing best.
- Applied SMOTE method to handle class imbalance along image preprocessing and feature extraction to improve prediction quality.

Predictive Maintenance in Railways (Python, LSTM, TensorFlow, ReactJS)

- Built LSTM-based predictive maintenance system on 8,000+ multi-sensor time-series data, achieving 87.3% accuracy and 89.4% ROC AUC (10.5% better than traditional ML).
- Developed a ReactJS real-time dashboard with dynamic visualizations and failure probability alerts for live monitoring.

YouTube Transcript Summarizer – Flask TextRank NLP App (HTML, CSS, JavaScript, Python, Flask)

- Built a Flask app that fetches YouTube transcripts and generates TLDRs with TextRank; delivers adjustable summaries in seconds, reducing time-to-insight by 80–90% on typical 10–20 minute videos.
- Added robust error handling for videos without transcripts and a clean HTML, CSS, and JavaScript UI.

Crowdfunding & Event Management Platform (ReactJS, Django, MySQL)

- Designed a web platform for technical fest registrations and payments, enabling students to browse events, register, and pay securely online through Razorpay.
- Built an admin dashboard for committee to manage participants, approve campaigns, and track funds with real-time progress and fraud prevention.

CERTIFICATES

Data Base Management System: NPTEL (2023)

Linux OS: Spoken Tutorial (2024)

Python: Spoken Tutorial (2023)

Data Analytics and Visualization: Accenture (Forage-2024)

CO-CURRICULAR ACTIVITIES

- **CEO (IoT Cell):** Organized IoT events, workshops, and hackathons while managing team coordination and event execution.